ENGINEERS WALLAH

CHEMISTRY

ENGINEERS WALLAH

Lecture X

Acids, Bases and Salts



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TOPICS *to be covered*

1. Types of Acids on Basis of Source
2. Acid-Base Indicators
3. Chemical Properties of Acids and Bases
4. pH and Its Importance in Daily Life
5. Salts in One Shot

Knowledge Ride On



Types of Acids on Basis of Source

Knowledge Ride On



Acid-Base Indicators

Knowledge Ride On



Chemical Properties of Acids and Bases

Knowledge Ride On



pH and Its Importance in Daily Life

Knowledge Ride On



Salts in One Shot

Knowledge Ride On



Efficiency Hacks

Glossary

Acids (अम्ल)
Bases (क्षार)
Salts (लवण)
Indicator (सूचक)
Olfactory indicator (घ्राण सूचक)
Displacement reaction (विस्थापन प्रतिक्रिया)
Neutralisation reaction (उदासीनीकरण प्रतिक्रिया)

Concept Bites – Types of Acids on Basis of Source

Types of Acids on Basis of Source

Vicinity of nettle plant → Dock Plant (Basic in nature)

ORGANIC ACID – Living Sources – Plants & Animals

I

Natural source	Acid	Natural source	Acid
Citrus fruits	Citric acid	Nettle sting	Formic acid / Methanoic acid
Tamarind	Tartaric acid	Apple	Malic acid
Tomato	Oxalic acid	Ant sting	Formic acid
Vinegar (सिरका)	Acetic acid	Curd	Lactic acid

lemon, orange etc. → VITAMIN C (Ascorbic acid)

II

Mineral or inorganic acids are obtained from rocks or minerals. (non-living sources)

Example: H_2SO_4, H_2CO_3, HNO_3, HCl etc.

H_2SO_4 → Sulphuric acid; H_2CO_3 → Carbonic acid; HNO_3 → Nitric acid; HCl → Hydrochloric acid

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Which of the following acid is present in sour milk?

RRB JE 2014

A. Glycolic acid

B. Lactic acid ✓

C. Citrus acid

D. Tartaric acid

Milkman
↓
Raw milk | Fresh milk
↓
Lactic acid forming bacteria
↓
Acidic character (↑)
↓
SOURING OF MILK
(smell, taste & texture of milk)

To prevent this ←

Milkman adds baking soda (basic) to milk.

Concept Bites – Acid-Base Indicators



Give a Thought

To identify that a substance is an acid or a base, tasting it is the best possible option.

Acid - Generally, 'sour' ✓

Base - Bitter ✓

- Slippery to touch (Texture) ✓



Give a Thought

To identify that a substance is an acid or a base, tasting it is the best possible option.

(सलाह)

(i) *No! This can be disastrous as some acids and bases are highly corrosive, i.e. they produce a burning sensation on the skin and damage it.*

(ii) *Acids can taste bitter sweet as well as sour. So, tasting is not a relied option.*

Indicators that Show a Change in Colour

(सूचक)

Indicator	Neutral solution	Acidic solution	Basic solution
(a) Litmus solution	Pale Purple / Mauve	Red	Blue
(b) Turmeric paste	Yellow	Yellow	Brick Red
(c) Hydrangea flowers	Blue	Blue	Pink
(d) Phenolphthalein	Colourless	Colourless	Pink
(e) Methyl orange	Orange	Red	Yellow

Nature ← (a), (b), (c)

Synthetic ← (d), (e)

Indicators that Show a Change in Odour

'Smell'

(घ्राण सूचक)

- Olfactory indicators

'helpful for visually impaired people'

Olfactory indicator	Acidic solution	Basic solution
(a) Onion odoured cloth strips	Smell remains same	Smell vanishes
(b) Vanilla essence	Smell remains same	Smell vanishes
(c) Clove oil	Smell remains same	Smell vanishes

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

How do you know that vinegar is an acid?

RRB JE 2015

A. It will turn blue litmus red.

B. It will turn red litmus blue.

C. Phenolphthalein turns pink when added to vinegar.

D. It's pH will be more than 8.

Question

A blue litmus will turn _ _ _ _ _ when contacted with acid.

SSC JE 2020

A Blue

B Black

C Orange

D Red

Question

Which of the following act as an olfactory indicator?

RRB JE 2015

Colour change Odour change (Olfactory)

A Phenolphthalein (✓)

B Methyl orange (✓)

C Vanilla essence (✗) (✓)

D Litmus solution (✓)

Concept Bites – Chemical Properties of Acids and Bases

Question

A gas is evolved when a piece of zinc metal is placed in dilute sulphuric acid (H_2SO_4) what is the gas?

Displacement Rxn

Metal + dil. acid → Salt + Hydrogen gas

$Zn + dil.H_2SO_4 \rightarrow ZnSO_4 + H_2$

Squeaky pop sound (small explosion)

K	Kudi	Potassium
Na	Naal	Sodium
Ca	Car	Calcium
Mg	Mango	Magnesium
Al	Alto	Aluminium
Zn	Zisko	Zinc
Fe	Fir	Iron
Pb	lekar	Lead
H	thum	Hydrogen
Cu	Chale	Copper
Hg	Mathura	Mercury
Ag	sath	Silver
Au	Ghumne	Gold

Pt Rahek Platinum

A. Hydrogen

B. Oxygen

C. Water-vapour

D. Sulphur dioxide

Question

egg shell → Calcium carbonate ($CaCO_3$)

$H_2CO_3 \rightarrow H_2O + CO_2$

When we put a raw egg inside a beaker containing vinegar, there are bubbles of gas (X). Identify the gas (X).

(सिरका)

A H_2

B CO

C CO_2 ✓

D H_2S

Double displacement rxn
- Gas forming rxn

Metal carbonate/ hydrogen carbonate	+	Acid	→	Salt	+	Water	+	Carbon dioxide

$CaCO_3(s) + CH_3COOH(aq) \rightarrow Ca(CH_3COO)_2(aq) + H_2O(l) + CO_2(\uparrow)$

C A A C

Calcium acetate

Turns limewater milky

Double displacement rxn
- Precipitation Rxn

Limewater — insoluble

$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s) + H_2O(l)$

'Base' 'Acid' 'Salt'

Solution turns TURBID/milky

Excess of carbon dioxide gas passed through limewater

'clear solution'

$CaCO_3(s) + H_2O(l) + CO_2(g) \rightarrow Ca(HCO_3)_2(aq)$

'soluble'

Question

Neutralisation Rxn → Type of double displacement rxn

A black metal oxide (X) reacts with dilute hydrochloric acid to form another compound (Y) that turns the colour of the solution to blue-green. Identify (X), (Y) and the type of the reaction.

CuO

$CuCl_2$

'Neutralisation Rxn'

BASE + ACID → SALT + WATER

Metal oxide	+	Acid	→	Salt	+	Water
CuO(s)	+	2HCl(aq)	→	$CuCl_2$(aq)	+	H_2O(l)

Black oxide

Metal oxides, metal hydroxides and ammonium hydroxide are basic in nature.

A (X): Fe_3O_4, (Y): $FeCl_2$, Double displacement

B (X): Cu_2O, (Y): $CuCl_2$, Neutralisation

C ✓ (X): CuO, (Y): $CuCl_2$, Neutralisation

D (X): FeO, (Y): $FeCl_2$, Neutralisation

Question

What are the products of neutralization reaction?

RRB JE 2019

A Base and Salt

B Acid and Base

C Salt and Water

D Acid and Salt

Concept Bites – pH and its Importance in Daily Life

'power of Hydrogen ion'

pH Scale

Acids release → $H^+(aq)$ ion

Bases release → $OH^-(aq)$ ion

room temp.

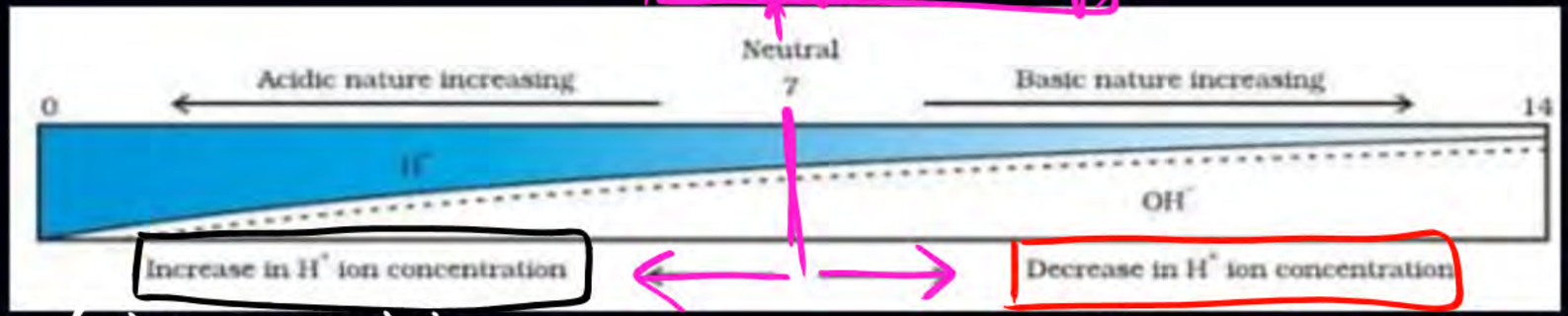
At 25 °C, the solutions may have pH from 0 to 14 For different solutions

the pH values will be:

- Neutral solutions: pH = 7
- Acidic solutions: pH < 7
- Basic solutions: pH > 7

Concentration

$H^+(aq) = OH^-(aq)$



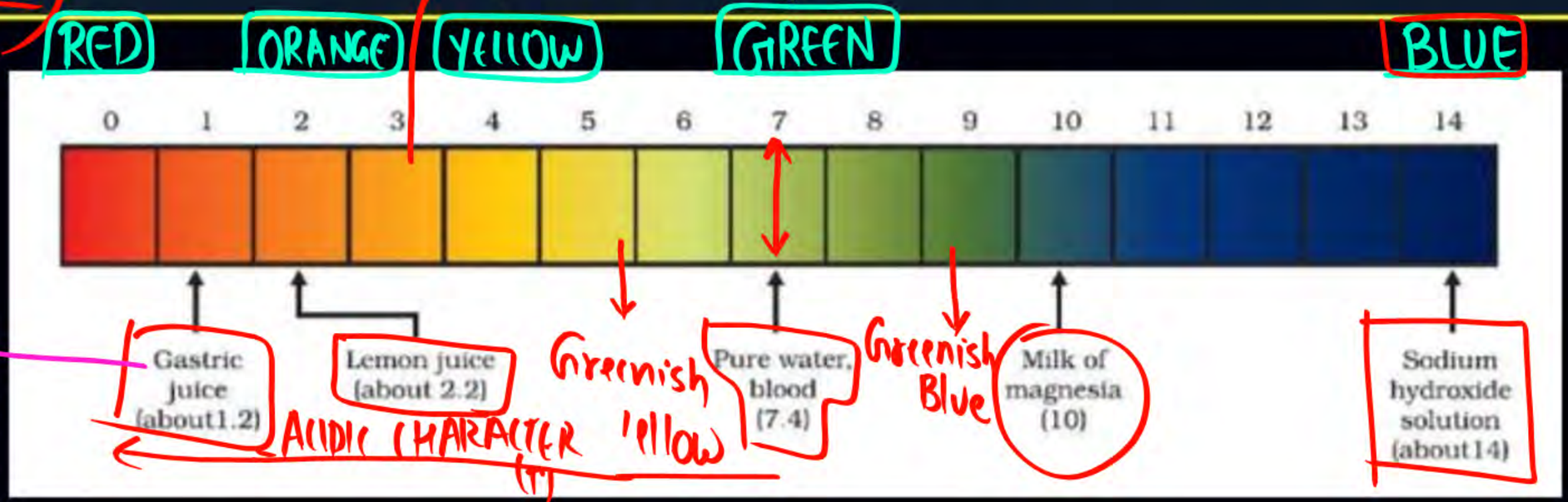
(↓) in $OH^-(aq)$ ion

(↑) in $OH^-(aq)$ ion

Universal Indicator

The pH of a solution can be measured using a universal indicator which is a mixture of thymol blue, phenolphthalein etc.

It shows different colours at different H^+ ions concentration in the solution.



- HCl
- lipase
- pepsin

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

Where will you place vinegar if you were to show its pH on a scale showing pH from 1 to 8 to?

RRB JE 2015

A Left of 7

B Right of 7

C On 7

D On 8

Question

The pH of water is:

RRB JE 2015

A 5

B 6

C 7

D 8

Question

The burning sensation of a bee sting can be stopped by rubbing the affected area with soap. This is because

'Neutralisation Rxn'

(base)

A a bee sting is acidic and soap, an alkali, neutralises it

'formic acid | methanoic acid'

B a bee sting is alkaline and soap, an acid, neutralises it

C soap cleans the affected area and removes the sting

D soap acts as an anaesthetic and dulls the sensation

WASP STING
↓
Alkaline (Basic)
↓
To neutralise it apply
- lemon juice
- vinegar

Question

Which among the following is should be taken to get relief from acidity?

- **A** Vinegar (X)
- **B** Lemon juice (X)
- **C** Milk of magnesia $Mg(OH)_2$ → pH = 10 ✓
- **D** Citrus fruits (X)

- overeating
- spicy food
- caffeinated drinks

→ more gastric juice is released than required

→ Acidic

→ HCl

Question

The tooth enamel is made from which of the following?

A. Magnesium carbonate

B. Calcium phosphate $Ca_3(PO_4)_2$

C. Magnesium phosphate

D. Sodium carbonate



→ Tooth enamel
- hardest part
- doesn't regrow

pH falls below 5.5 → Tooth cavity

→ neutralised by

Toothpaste → basic in nature

Question

Which among the following acid is not present in acid rain?

(अम्ल वर्षा)

A. Hydrochloric acid ✗

B. Nitric acid ✓

C. Carbonic acid ✓

D. Sulphuric acid ✓

$CO_2 + H_2O \rightarrow H_2CO_3$

$NO_2 + H_2O \rightarrow HNO_2 + HNO_3$

$HNO_2 + O_2 \rightarrow HNO_3$

$SO_2 + H_2O \rightarrow H_2SO_3$

$H_2SO_3 + O_2 \rightarrow H_2SO_4$

Concept Bites – 'Salts' in One Shot

Ionic Compound

Sodium hydroxide

alkali - bases soluble in water

Common name: → Caustic soda

Chemical Formula: → NaOH

Manufactured by: → Chlor-alkali process

ex: Na_2O, NaOH, K_2O, KOH etc

'CHLORINE' ← by-product (secondary product) ← Chlor

'NaOH' (main product) ← alkali

Electrolysis of brine solution (NaCl + H_2O)

$$2NaCl(aq) + 2H_2O(l) \xrightarrow{\text{Electricity}} 2NaOH(aq) + Cl_2(g) + H_2(g)$$

Cl_2 → By-product

Calcium oxychloride

Chemical Formula:	$CaOCl_2$ (actual formula is complex)
Common Name:	Bleaching Powder
Manufactured by:	Hasenclever Plant & Beckmann Plant
Principle of Manufacture	The chlorine gas obtained as a by-product from the chlor-alkali process when reacts with dry slaked lime [$Ca(OH)_2$] produces bleaching powder.

$$Ca(OH)_2(s) + Cl_2(g) \rightarrow CaOCl_2(s) + H_2O(g)$$

Common Name →	Soda Ash	Baking soda
Chemical Formula: →	Na_2CO_3	$NaHCO_3$
Chemical Name: →	Sodium carbonate	Sodium bicarbonate
Manufactured by: →	Ammonia-soda process / Solvay's Process	

$$NaCl + H_2O + CO_2 + NH_3 \rightarrow \underset{\text{(Ammonium chloride)}}{NH_4Cl} + \underset{\text{(Sodium hydrogencarbonate)}}{NaHCO_3}$$

→ INTERMEDIATE PRODUCT

Thermal decomposition →

$$\underset{\text{(Sodium hydrogencarbonate)}}{2NaHCO_3} \xrightarrow{\text{Heat}} \underset{\text{(Sodium carbonate)}}{Na_2CO_3} + H_2O + CO_2$$

→ MAIN PRODUCT

Imp.

Na_2CO_3

↓ one of its form

$Na_2CO_3 \cdot 10H_2O$

- WASHING SODA
- Sodium carbonate decahydrate

Baking Powder

↓

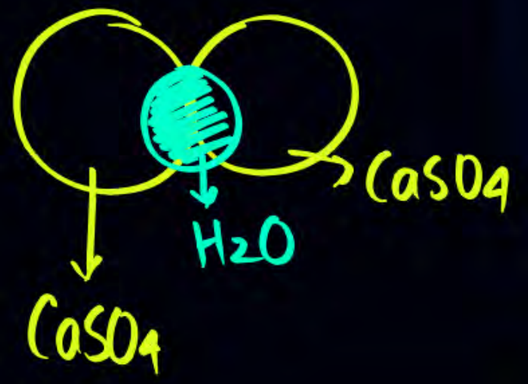
Baking Soda + Cornstarch + Cream of tartar (Tartaric acid)

dry → acid

Plaster of Paris

Chemical Formula: → $CaSO_4 \cdot \frac{1}{2} H_2O$ or $(CaSO_4)_2 \cdot H_2O$

Chemical Name: → Calcium sulphate hemihydrate



Prepared by: --→ When gypsum ($CaSO_4.2H_2O$) is heated in a kiln at 100 °C or 373 K, it loses three-fourth of its water of crystallisation forming the plaster of Paris.

$$CaSO_4 \cdot 2H_2O(s) \xrightarrow{T = 100^\circ C} \underbrace{CaSO_4 \cdot \tfrac{1}{2} H_2O(s)}_{\text{P.O.P}} + 1\tfrac{1}{2} H_2O(g)$$

- Gypsum
- Calcium sulphate dihydrate

Plaster of Paris

Dead Burnt Plaster:

$CaSO_4.2H_2O \xrightarrow{T > 100^\circ C} CaSO_4 + 2H_2O$

$CaSO_4$ → Dead burnt plaster

Setting of PoP: → When plaster of Paris is mixed with water and left for half an hour to one hour, it rehydrates to form a hard mass, i.e. gypsum.

PoP → Gypsum

$$CaSO_4.\frac{1}{2}H_2O(s) + 1\frac{1}{2}H_2O\ (l) \rightarrow CaSO_4.2H_2O(s)$$

Practice and PYQ's Wallah



PW Ka ChemStar!

Question

What is the chemical name of the compound $CaOCl_2$?

RRB JE 2019

A Washing soda

B Baking soda

C Bleaching powder

D Tartaric acid

Question

Which salt of calcium is Plaster of Paris?

RRB JE 2015

$CaSO_4 \cdot \frac{1}{2} H_2O$

'Calcium sulphate hemihydrate'

A Calcium sulphate (Dead burnt plaster)

B Calcium oxide

C Calcium carbonate

D Calcium hydroxide

Efficiency Hacks!

Pomodoro Technique

- Identify a task or tasks that you need to complete.
- Set a timer for 25 minutes.
- Work on a task with no distractions.
- When the alarm sounds, take a 5-minute break.
- Repeat the process 3 more times.
- Take a longer 30-minute break and start again.




#sbsathhai
#pwsathhai
Thank
You !